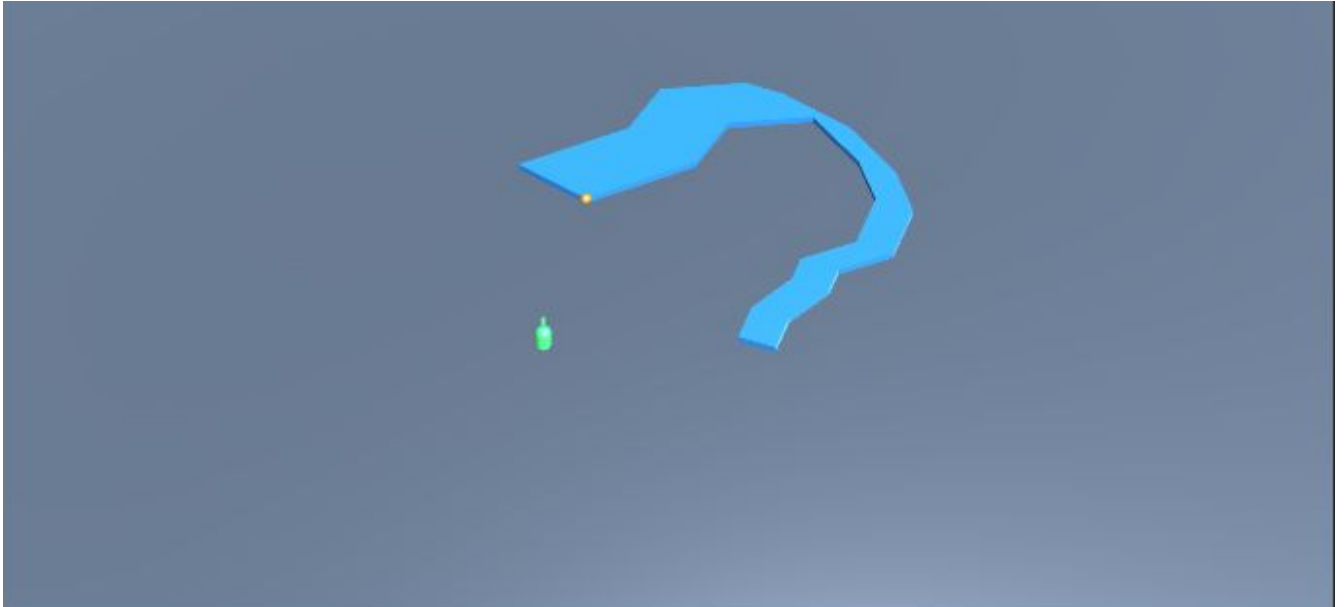


Volumetric Audio

Unity Audio, FMOD and Wwise - three ways to play the same volumetric sound.



Example scene: blue sound volume, green listener and orange emitter.

Choose your audio backend

Backend	Where to start
Unity Audio	No additional audio SDK required. Installation: page 2. Setup: pages 3-4.
FMOD	Install FMOD for Unity and FMOD Support. Core example: page 5. Your own Studio events: pages 6-7.
Wwise	Install Wwise Unity Integration and Wwise Support. Event and SoundBank setup: page 8. Unity components: page 9.

Volumetric Audio moves a sound emitter to the point on a volume closest to the listener. A long river or a large area can therefore sound like an extended source instead of a single speaker at its center.

Keep the volume and emitter separate

The **volume object** holds the mesh. The **emitter object** moves during playback. Use a different Transform for each one.

Screenshots show the actual components in Unity 6.3 LTS. Field names match the Inspector so you can find them easily. Troubleshooting: page 10.

Import the backend you need

01 Install the Unity dependencies

Open **Window > Package Manager** and install Burst, Collections and Mathematics. If needed, use **Install package by name**: com.unity.burst, com.unity.collections and com.unity.mathematics. Versions 1.8.28, 2.6.2 and 1.3.2 were tested locally with Unity 6000.3.18f1.

02 Import the base package

Choose **Assets > Import Package > Custom Package**, select **VolumetricAudio.unitypackage** and import its contents. The package appears under Assets/VolumetricAudio.

03 Install an optional audio backend

For FMOD or Wwise, install the official Unity integration first. Then double-click the matching nested package in **VolumetricAudio/Integrations** and click Import. Skip this step when using Unity Audio.

File in Integrations	Added folders
FMOD Support.unitypackage	Integrations/FMOD and Examples/FMOD
Wwise Support.unitypackage	Integrations/Wwise and Examples/Wwise

Open an example scene

Each backend has a Scenes folder inside Examples.

Scene in Examples	How to run it
UnityAudio / UnityAudio.unity	Open the scene and press Play.
FMOD / FMOD.unity	Install the SDK and add-on, then press Play. The Core example does not need banks.
Wwise / Wwise.unity	Configure initialization and banks first. In Play Mode, enter an event name and click Play in the WWISE EVENT panel.

Shared example controls

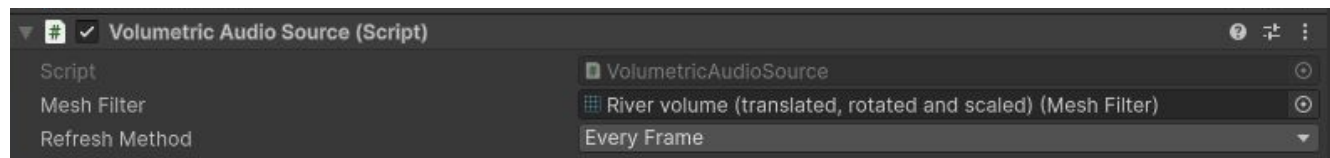
WASD / arrow keys: move the listener; **Q / E**: turn; **Shift**: move faster; **Tab**: point / volume; **Space**: automatic route; **M**: mute this example. In Wwise, M stops and restarts the emitter's event.

If keyboard input does not work

Examples use the classic Unity Input API. If your project uses only the new Input System, set **Active Input Handling = Both** in Player settings. Unity may require an editor restart.

One shared listener, one Mesh Filter

Use the same setup for Unity Audio, FMOD and Wwise.



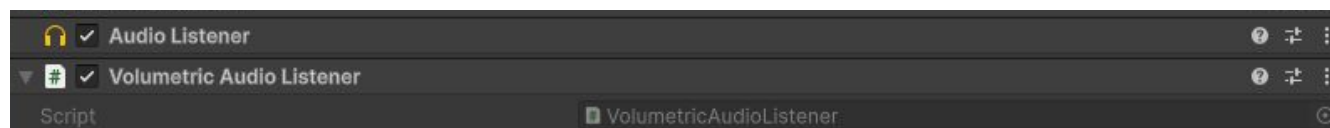
Volumetric Audio Source: assign the volume object's Mesh Filter and choose a refresh method.

01 Create the volume object

Add an object with a **Mesh Filter** and a mesh describing the sound area. For an imported model, enable **Read/Write** in its Model import settings. The volume object stays separate from the moving emitter.

02 Register the shared listener

Add **Volumetric Audio Listener** to your listener object using Add Component. Keep exactly one enabled component. Every source reads **VolumetricAudioListener.Instance**; no source has a listener reference field. Add the backend listener to this same object: Audio Listener, FMOD Studio Listener or AkAudioListener.



Unity Audio listener object: Audio Listener plus the shared Volumetric Audio Listener.

03 Add and configure the emitter

Drag the appropriate emitter prefab into the scene. Set **Mesh Filter** to the separate volume's Mesh Filter; the source reads both **sharedMesh** and **transform** from it. Choose **Refresh Method = Every Frame** for the first test.

Field	What to assign
Mesh Filter	The MeshFilter on the volume object, such as River volume. Do not assign the emitter's decorative sphere.
Refresh Method	Every Frame: continuous updates. Every4thFrame: less frequent updates. Manual: call RefreshPosition() from your own code.

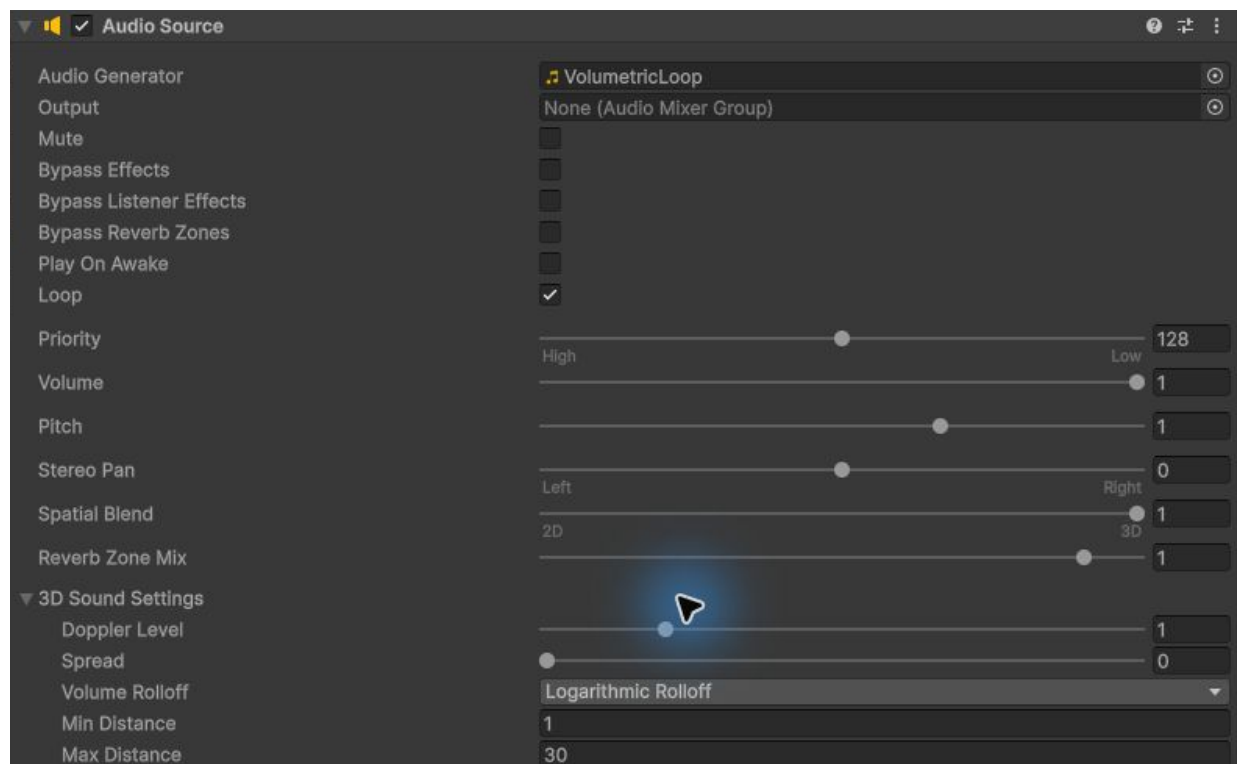
Changing geometry and listeners

Moving, rotating or scaling the Mesh Filter, or replacing its sharedMesh, is detected on the next refresh. After editing vertices of the same mesh at runtime, call **RefreshMesh()**. To change listeners, disable the old Volumetric Audio Listener before enabling the replacement. Without an active listener, sources keep their last position.

Open meshes provide the closest surface point. Interior detection requires a valid closed mesh. The package positions emitters; acoustic occlusion and reverb require your own audio setup.

Use a standard Audio Source

Prefab: Prefabs/VolumetricAudioSourcePrefab.prefab. Listener: the standard Unity Audio Listener.



Example settings: Spatial Blend = 1, Loop enabled, Min Distance = 1 and Max Distance = 30.

01 Assign an audio clip

In **Audio Source**, assign your AudioClip. In Unity 6.3 the field is called **Audio Generator**; older editors may label it AudioClip. For a quick test, use Examples/Shared/VolumetricLoop.wav.

02 Enable spatial playback and looping

Set **Spatial Blend = 1 (3D)**, Volume = 1 and enable Loop. Expand **3D Sound Settings**: start with Min Distance = 1 and Max Distance = 30. Adjust the distances to your scene scale.

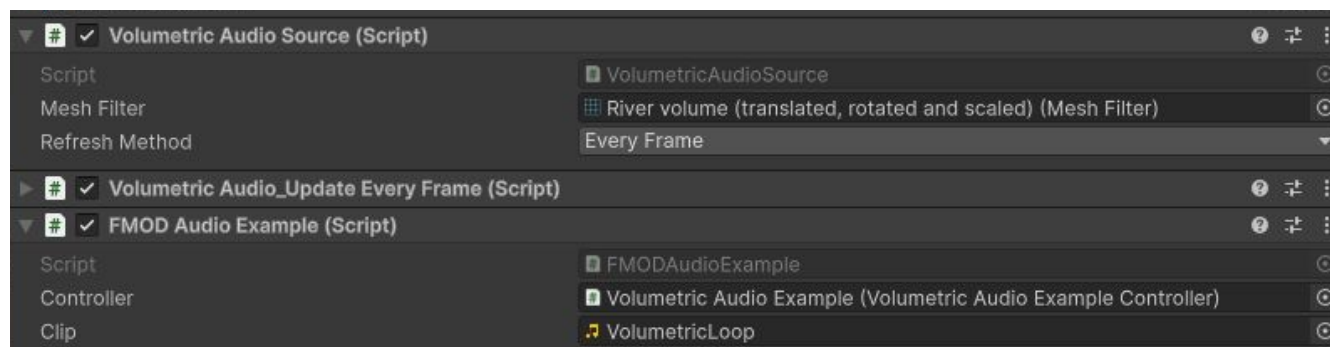
03 Set playback and the listener

In your own scene, enable **Play On Awake** or call AudioSource.Play() from code. Keep one active Audio Listener and add Volumetric Audio Listener to the same object. In the example, Play On Awake is disabled because Unity Audio Example starts playback.

Check: approach the river, move away and press Tab. The emitter should follow the closest point on the volume, and the sound should become quieter with distance.

Test FMOD without sound banks

Scene: Examples/FMOD/Scenes/FMOD.unity. This example plays the included loop through FMOD Core 3D.



FMOD Audio Example references the shared Controller and the VolumetricLoop clip. This scene does not use Studio Event Emitter.

01 Install the integration

Install the official **FMOD for Unity** integration. Version 2.02.09 was tested locally. Then import **Integrations/FMOD Support.unitypackage**.

02 Open the FMOD scene

After compilation finishes, open FMOD.unity. The shared listener, volume Mesh Filter and clip are already configured. Press Play, then use WASD and Tab to compare point and volumetric playback.

03 Replace the test clip if needed

Assign an AudioClip in **FMOD Audio Example > Clip**. Set its import setting **Load Type = Decompress On Load** so the example can read the samples. Keep Controller assigned to the example controller in this scene.

Core example versus Studio integration

This scene creates and releases its own FMOD Core system and sends listener positions directly, without Studio Listener. It does not need an event, banks or a Studio project. For events in your game, use the Studio prefab described on the next two pages.

If the example is silent

Check that FMOD native libraries are installed for the active platform, that your output device is audible and that Clip is assigned. Do not add another Unity AudioSource to play the same loop in this scene. In Play Mode, the orange emitter should move with the volume calculations.

Prepare your own 3D event

01 Create a looping event

In your FMOD Studio project, create an event such as **event:/Environment/VolumetricLoop**. Add an audio file to a track and configure continuous playback using a loop region or looping instrument.

02 Enable spatial sound

Add **FMOD Spatializer** to the event's master track. For an initial test, use Min Distance = 1 and Max Distance = 30. A purely 2D event will not make the emitter's movement audible.

03 Build a bank

Assign the event to a bank and build the banks in FMOD Studio for your target platform. Use a Studio version compatible with your Unity integration.

04 Select the project or banks in Unity

In **FMOD settings**, configure the bank source for your workflow: a Studio project or previously built banks. Refresh the events in FMOD Event Browser.

05 Load the bank before playback

For a simple test, use FMOD's bank loading at initialization. If you manage banks manually, use **Studio Bank Loader** or your own manager and wait for the bank to finish loading before starting the event.

Is the event ready?

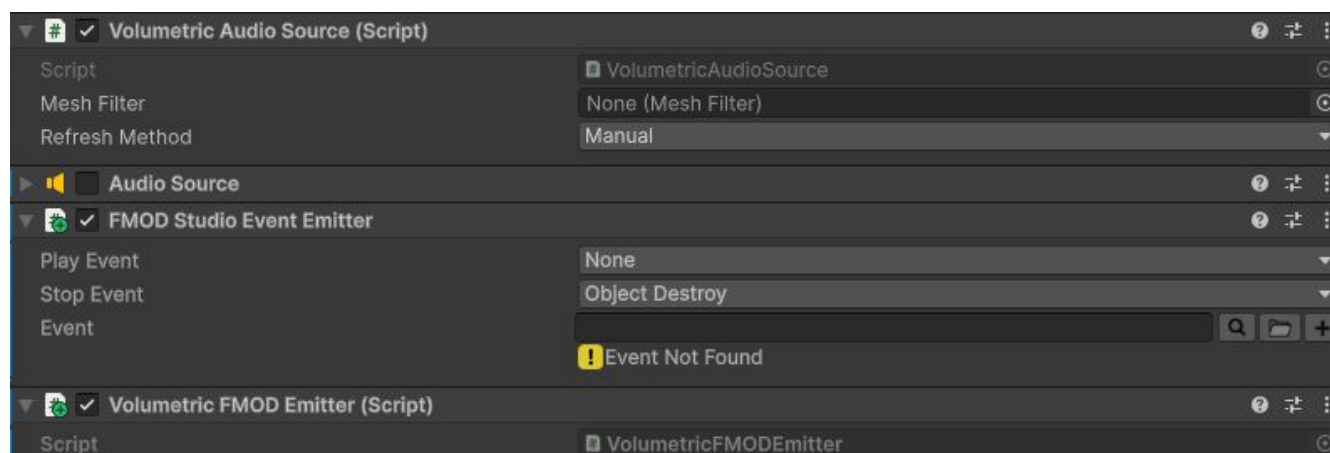
Audition the event in FMOD Studio first. Then confirm that its path appears in Unity and that its bank is loaded. Once those checks pass, configure the Volumetric Audio emitter.

Which tool controls which settings?

FMOD Studio	Unity
Audio file, looping, Spatializer and attenuation curve	Volume mesh, listener, emitter prefab, selected event and playback trigger
Bank contents and generation	Bank source and loading before playback

Component and bank-loading reference: [FMOD for Unity - Game Components](#). Setting names may vary between integration versions.

Connect the prefab to your event



Unconfigured prefab: assign the volume Mesh Filter and event. The shared listener is configured once on its own object.

01 Add the prefab to your scene

Use **Integrations/FMOD/Prefabs/VolumetricAudioSourcePrefab_FMOD Variant.prefab**. Assign the volume Mesh Filter as described on page 3; choose Every Frame. Keep the inherited Unity Audio Source disabled.

02 Add an FMOD listener

Add **FMOD Studio Listener** and **Volumetric Audio Listener** to the same camera or player object. Each volumetric source uses the static Instance automatically. There is no per-emitter listener field.

03 Select the event and playback trigger

Select your 3D event in **FMOD Studio Event Emitter > Event**. For a simple test, set **Play Event = Object Start** and **Stop Event = Object Destroy**; the bank must already be loaded. The default Play Event value, `None`, means playback starts from your own code, for example `StudioEventEmitter.Play()`.

04 Test playback

Press Play. Move the listener along the volume and away from it. Volumetric FMOD Emitter forwards the final emitter position to the active event in `LateUpdate`.

One active volumetric listener

All volumetric emitters use the same enabled Volumetric Audio Listener. If your FMOD project has several Studio listeners, configure its listener routing to match this object. To switch the shared listener, disable the old component before enabling the new one.

Set up the event and SoundBank

01 Install and connect Wwise

Install the official **Wwise Unity Integration** using Audiokinetic Launcher and connect Unity to your Wwise project. Then import **Integrations/Wwise Support.unitypackage**.

02 Create a looping Sound SFX

In Wwise, import **Examples/Shared/VolumetricLoop.wav** or your own audio file into a Sound SFX under Containers (Actor-Mixer Hierarchy in older Wwise versions). Enable looping. Start with one simple sound for the first test.

03 Configure positioning

In **Positioning**, use emitter positioning, enable **Listener Relative Routing** and set **3D Spatialization** to Position or Position + Orientation. Assign Attenuation with a volume curve: full volume near 1 m and silence around 30 m. If these settings are inherited, enable their override on the selected object first.

04 Create a Play event

Create an event named **Play_VolumetricLoop** with a Play action targeting that Sound SFX. You can use a different name, but Unity must use exactly the same event name.

05 Generate the banks

Add the event and its media to a SoundBank, such as **VolumetricExample**. Generate banks for the test platform and refresh them in the Unity integration. Init and the event bank must come from the same Wwise project.

06 Configure initialization and loading

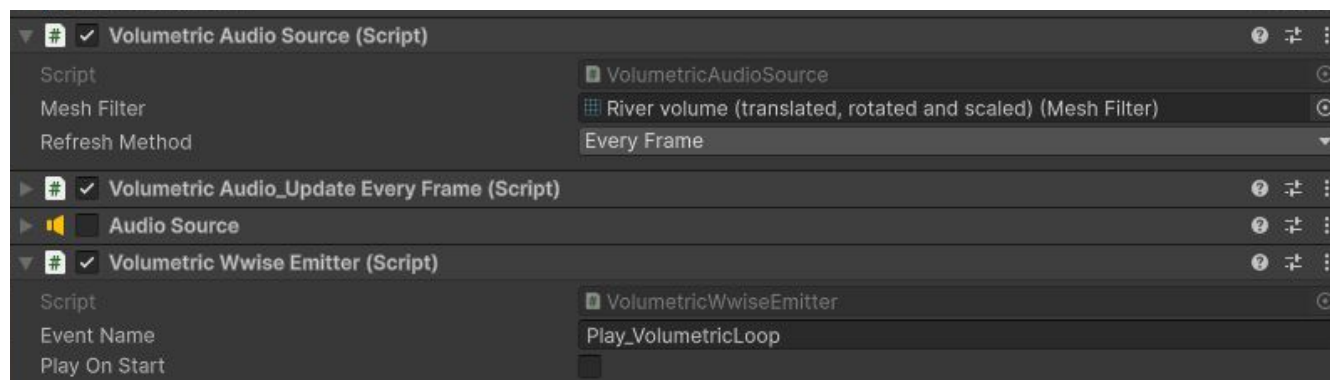
Use the integration's standard **WwiseGlobal** configuration. Load Init and the event SoundBank before clicking Play in the example panel. Use the official bank workflow, such as AkBank or your project's bank manager. A .bnk file being present does not mean it has been loaded.

Validated Wwise configuration

Native playback was validated with **Wwise 2025.1.11 Build 9262** and **Unity 6.3 LTS 6000.3.18f1** in **Unity Editor Play Mode** and a **Windows x64 Mono player**. Checks cover shared-listener routing, moving and transformed meshes, distance attenuation, mute, stop and restart. Screenshots on the next page show this package's components.

References: [Audiokinetic - Managing SoundBanks](#) and [3D positioning illustrated](#).

Configure the emitter and play



Emitter components in the Wwise scene. Unity Audio Source is disabled; the event name is `Play_VolumetricLoop`.

01 Use the example scene

Open **Examples/Wwise/Scenes/Wwise.unity** and configure WwiseGlobal and banks as described on page 8. The volume Mesh Filter and shared Volumetric Audio Listener are configured. After Wwise initializes, the example adds AkAudioListener to this same listener object.

02 Or add the prefab to your own scene

Use **Integrations/Wwise/Prefabs/VolumetricAudioSourcePrefab_Wwise.prefab**. Assign the volume Mesh Filter and refresh method from page 3. Add Volumetric Audio Listener and AkAudioListener to the same listener object; AkAudioListener requires AkGameObj. The adapter adds AkGameObj to the emitter after the Wwise engine is ready.

Volumetric Wwise Emitter	Setting
Event Name	The Play event name from a loaded SoundBank, such as <code>Play_VolumetricLoop</code> .
Play On Start	Your own scene: enable if the bank is ready before playback starts. Example: disabled because you start playback using the panel.

03 Play and compare

In Play Mode, enter the event name in **WWISE EVENT** and click Play. Stop affects only this emitter's event. Tab switches point / volume mode; M stops and restarts this event.

Missing SDK? The panel displays installation instructions. **Event failed?** Check the event name, initialization, loaded bank and Wwise Profiler messages. Check that AkAudioListener is on the object with the active Volumetric Audio Listener. Keep Unity Audio Source disabled.

If something does not work

Symptom	First checks
Errors after importing the base package	Are Burst, Collections and Mathematics installed? The base package does not require FMOD or Wwise SDKs.
The emitter does not move	Assign the volume Mesh Filter and enable one Volumetric Audio Listener. Change Manual to Every Frame. Enable mesh Read/Write.
The river or entire volume moves	Mesh Filter points to the moving emitter. Assign the Mesh Filter on a separate volume object.
The sound remains 2D	Unity: Spatial Blend = 1. FMOD: Spatializer in the event. Wwise: 3D Spatialization and Listener Relative Routing.
Unity Audio is silent	Check the clip, Volume, Mute, active Audio Listener and playback start. The example starts from a script; use Play On Awake or Play() in your own scene.
FMOD: Event Not Found	Expected on a prefab with no event assigned. Select an event after building and refreshing the banks. Check that its bank is loaded.
Wwise: not ready / Event failed	Check the official SDK, WwiseGlobal, Init, event bank and exact event name. The adapter waits up to 10 seconds for initialization; restart Play Mode after fixing the setup.
Keyboard controls do not work	Focus the Game view. If using the new Input System, check Active Input Handling = Both.
The sound plays twice	Use one audio backend per emitter. Keep Unity Audio Source disabled for FMOD and Wwise.

Quick verification checklist

1. The emitter moves to the point on the volume closest to the listener.
2. Moving away reduces volume; turning the listener changes the sound direction.
3. Tab in the example gives an audible point / volume comparison.
4. Stop or exiting Play Mode ends the example's playback.
5. Console reports no missing references, unreadable meshes or bank errors.

Verified locally: Unity 6.3 LTS 6000.3.18f1; Unity Audio and FMOD Core 2.02.09 playback; native Wwise 2025.1.11 Build 9262 playback in Unity Editor Play Mode and a Windows x64 Mono player; base import without audio SDKs; shared-listener lifecycle and transformed Mesh Filter calculations. Native audio capture confirmed audible playback near the volume and silence at distance and while muted.